

REAL BUILDS. MEASURABLE OUTCOMES.

Case Studies & Technical Pedigree

— From hackathon submissions to healthcare infrastructure —

Full-Stack Engineering

Next.js · React · TypeScript ·
FastAPI · PostgreSQL

Autonomous AI Workflows

LLM orchestration · RAG
pipelines · MCP systems

Robotic Process Automation

Playwright headless bots at
computational speed

Transparent Pricing

Published rates. Free working
prototype before any payment
commitment.

Engineering-First Culture

The founder writes production
code personally on every client
project.

Deterministic AI

Strictly typed scaffolding
enforced around every
probabilistic LLM call.

Lighthouse 96

HearWise performance score

Production deployment on Vercel

Meta Hackathon

OpenEnv-Debugger submitted

Global AI competition recognition

100+

Portal types automated

Omni-Apply RPA bot architecture

COMPANY

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CASE STUDY 01

HearWise Child Health

Architecting Secure Healthcare Technology on Vercel & Supabase

THE CHALLENGE

Managing pediatric healthcare data requires handling highly sensitive PII across complex relational structures between parents, children, and practitioners. The platform demanded absolute data security while remaining intuitive for non-technical users on any device at any location.

TECHNOLOGY STACK DEPLOYED

- ◆ Next.js 14 (SSR + SSG hybrid)
- ◆ Supabase PostgreSQL + Auth
- ◆ Row Level Security (RLS) — SQL layer
- ◆ Vercel Edge Network + CI/CD
- ◆ TypeScript + Tailwind CSS
- ◆ Pydantic + FastAPI Backend

THE LIT ARCHITECTURAL SOLUTION

Frontend Architecture

A highly responsive, component-based Next.js architecture ensures zero-latency transitions and exceptional mobile performance for parents accessing the platform on the go. The dashboard was designed with clinical precision — reducing cognitive load for practitioners managing multiple child profiles simultaneously.

Database Security Architecture

The critical engineering achievement was implementing rigorous Row Level Security (RLS) policies directly into the Supabase SQL layer. A user's JWT token can only ever query rows associated with their specific UUID — mathematically eliminating the possibility of cross-tenant data leaks between patient families and practitioners.

Performance & Deployment

Deployed to production via Vercel for continuous integration and edge-network speed. The application achieved a Google Lighthouse Performance score of 96, ensuring fast, accessible healthcare data access from any device or connection speed — including low-bandwidth mobile connections in rural areas.

THE RESULT

A highly scalable, robust health-tech platform serving real users. The deterministic RLS architecture ensures absolute data integrity while the Vercel deployment delivers enterprise-grade performance. This project sets a new standard for open-source healthcare application architecture in India. Source available on GitHub.

CASE STUDY 02

OpenEnv-Debugger

Meta PyTorch Global Hackathon Submission

THE CHALLENGE

ML engineers and data scientists lose hundreds of hours annually fighting PyTorch environment conflicts, dependency hell, and cryptic terminal error logs. Manually reading massive stack traces and searching online for solutions creates a massive bottleneck that stalls entire teams. This is exactly the type of deterministic problem that LIT's engineering philosophy exists to eliminate permanently.

THE LIT ARCHITECTURAL SOLUTION

Python CLI Automation Core

A robust Command Line Interface tool actively monitors the developer's local environment. It captures complex stack traces, error logs, and environment state in real-time, producing a structured diagnostic payload before any AI call is made — ensuring the LLM receives clean, well-formatted context.

Deterministic AI Orchestration Layer

Instead of building a naive regex parser, we integrated advanced LLM logic. The script captures stack traces and pipes them into a strictly structured prompt, using AI to instantly diagnose the root conflict — for example, CUDA version mismatches or conflicting PyTorch and torchvision builds.

Actionable Output with Bash Commands

The tool does not just identify problems — it generates the exact bash commands required to resolve the environment state. The output is deterministic, copy-pasteable, and validated against known PyTorch environment patterns before being presented to the engineer.

RESULT & CORE PHILOSOPHY DEMONSTRATED

Successfully submitted to the Meta PyTorch Global Hackathon. This project perfectly demonstrates LIT's core philosophy: using deterministic software engineering (Python CLI) to harness probabilistic AI models (LLM) for practical, workflow-altering outcomes. The tool validates that AI is a component, not a solution.

CASE STUDY 03

Omni-Apply RPA Architecture

Autonomous Web Portal Navigation & Submission at Scale

Ongoing development of an autonomous application bot using DOM-scraping, Playwright, and LLM APIs to dynamically navigate, read, and auto-complete complex web portals. The system demonstrates the full three-pillar LIT automation stack operating in concert: Data Ingestion, Cognitive Parsing, and Autonomous Execution.

TECHNICAL ARCHITECTURE

◆ DOM Inspection Engine

Playwright captures the live HTML DOM structure of each portal. CSS selectors are dynamically identified per portal, making the bot resilient to minor layout changes and UI updates without requiring human reconfiguration.

◆ LLM Context Engine for Personalisation

Unstructured form questions are piped to the Claude API with the job description and resume as context. The AI generates hyper-personalised, on-brand responses in under 2 seconds, enforced by a strict JSON schema output structure.

◆ Parallel Execution via Python asyncio

Multiple Playwright browser instances run concurrently via Python asyncio, enabling simultaneous multi-portal submissions without interference between browser contexts or shared session state.

◆ Retry & Error Handling System

CAPTCHA detection triggers graceful fallback. Network timeouts trigger exponential backoff retry logic. Form validation errors trigger dynamic re-attempt with adjusted input. Zero manual intervention required.

COMMISSION YOUR OWN CASE STUDY

Logic Intelligence Technologies is accepting new engineering engagements for Q4 2025. We deliver a free working prototype before any payment commitment. Schedule a no-obligation technical discovery session via the website below.

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